FIN ST1722–ST1811

Repository-Wide Falsification of the Kernel-to-L_{total} Chain

and the Decisive Source–Action Gate

Krzysztof Żuchowski
Independent Researcher — Fractal Information Theory Project
ORCID: 0009-0002-0909-3613

Six adaptive research rounds — 90 programmes
Release 11.03 — 31 August 2026

This report audits the current FIN repository state, with particular attention to `STRICT_KERNEL_LAGRANGIAN_AND_EOM_DRAFT.md`, `STRICT_EQUATION_SHEET_KERNEL_TO_LTOTAL_CURRENT.md`, the strict/legacy split, APD completion, current no-go maps, and all recent strict and operational results. Its purpose is falsification, not confirmation.

English mathematical referee-style research report.

Abstract

ST1722–ST1811 perform an adversarial reconstruction of the current FIN route from legacy and strict kernels to effective moments, couplings, a displayed L_{total} and covariant equations of motion. The principal conclusion is negative but structurally decisive: the current chain is not a derivation of a unique gauge-covariant physical action.

First, the APD completion factor is analysed as a mathematical object. On the finite integer Z_{12} distance carrier, the legacy kernel is nonzero and the diagonal multiplier $Q_i = K_{\text{strict}}(i)/K_{\text{legacy}}(i)$ exists uniquely. But this is automatic for every nonzero input vector and arbitrary target. The multiplier has six positive and six negative values, so it is not a positive channel. In the real continuum the legacy cosine vanishes at $d = 4/3+4k$, while the strict target is nonzero at the first such points. Therefore the APD ratio has nonremovable poles and no bounded continuous global multiplicative completion exists. P2363 remains correct only as a finite- domain and local-germ moment identity; it is not an independently sourced global bridge.

Second, literal variation of the displayed action reveals internal defects. The term $-y\phi\bar{\psi}\psi$ produces $+y\bar{\psi}\psi$ in the scalar Euler row under the equation sheet's convention, whereas the compact draft displays the opposite sign. The interaction $A_\mu A^\mu \phi^2$ is not locally gauge invariant by itself and violates the off-shell Ward identity. A gauge repair requires a complex charged scalar with both derivative current and seagull terms tied to one charge, or an additional Higgs/Stueckelberg field. A Proca interpretation is possible only by abandoning the gauge/BRST claim. Moreover, the displayed covariant EOM rows contain Higgs, cubic, RF^2 and curvature- squared terms absent from the displayed action, while omitting explicit rows from its $A^2\phi^2$ interaction. No single literal displayed action generates all current covariant rows.

Third, the three-moment map at $d_{\text{ref}} = 1$ is neither reference invariant nor identifying. Its Jacobian with respect to $(\omega, \phi, \beta, \eta)$ has rank three, and distinct parameter tuples are constructed with the same (c_0, c_1, c_2) to numerical residuals near 10^{-12} but different kernel values away from the reference point. The effective multipliers change substantially when d_{ref} is moved, and five free base couplings absorb arbitrary effective targets wherever the factors are nonzero. A fourth jet locally restores parameter rank, but not physical scales or a source law.

Fourth, the finite strict kernel matrix W is tested as a propagator. In the current construction $W = sI - A$. It cannot equal a normal-ordered resolvent $a(A + m^2 I)^{-1} - cI$ on seven distinct eigenvalues. Its spectrum has seven negative and five positive modes, so using W as a Euclidean covariance gives an indefinite inverse Hessian. The unavoidable variational object is instead the positive Dirichlet action

$$S_A[f; J] = \frac{1}{2} f^T A f - J^T f,$$

whose Green operator is A^+ on the mean-zero subspace. This action unifies finite Green response, heat, unitary and wave functional calculi but does not select a physical channel, state category, scale or spacetime interpretation.

The current FIN mathematics therefore survives as a finite Dirichlet–Markov– spectral information geometry with conditional operational models. It is not currently a fundamental physical theory. The highest-value next theorem is to derive or refute one globally regular, refinement-compatible, gauge-covariant source action directly from strict algebra/Dirichlet data, with every EOM regenerated by literal variation. A seven-row decisive gate is provided; the current strict repository passes zero complete rows.

Contents

1 Repository scope and method

The audit uses the current project guardrails, strict/legacy split notes, strategic state maps, the two requested Lagrangian/EOM documents, P1866, P2316, P2362–P2366, the P2685–P2687 reverse-closure no-go sequence, the P3140–P3172 unit/selector state map, and Releases 10.89–11.02. The existing negative results are treated as established unless a stronger direct calculation changes their scope.

The six adaptive rounds were not precommitted. Each round selected the next object from the strongest defect found in the previous one:

action/EOM audit*o*APD*o*literal variation*o*moment map*o*propagator/action*o*decisive gate.

2 Round I: ST1722–ST1736 — action/EOM type audit

The displayed covariant action contains, schematically,

$$\{(\nabla\phi)^2, \phi^2, \phi^4, \phi^2 R, \bar{\psi} i D\psi, \phi\bar{\psi}\psi, F^2, A^2\phi^2, R, \Lambda\}.$$

The later covariant EOM rows additionally contain a separate H field, $\lambda_3\phi^2$, $\phi^2 H^\dagger H$, RH, RF^2, curvature-squared tensors and a full $T^{\mathrm{SM+mix}}_{\mu\nu}$. Those rows require additional action terms and field definitions; they cannot be generated by renaming the coefficients in the displayed scaffold.

Conversely, the later covariant E_ϕ and E_A rows do not explicitly carry the pair of terms generated by the displayed $A^2\phi^2$ interaction. The compact earlier scalar template does, showing that the two equation lists come from different lineages rather than one frozen action.

The compact statement also adds ordinary scalar/fermion/gauge Lagrangians to a gravity *density* already containing $\sqrt{-g}$. The later covariant presentation repairs that notation, but it does not repair the mixed term provenance.

Finally, solving two FRW residual equations algebraically for H^2 and Λ and substituting the solution back proves an algebraic normal form. It does not derive ρ, p, H, Λ, validate the off-FRW tensor equations or provide a physical background source.

3 Round II: ST1737–ST1751 — APD completion

3.1 Finite completion theorem

Let $L_i \neq 0$ on a finite carrier and let S_i be arbitrary. There is exactly one diagonal multiplier

$$Q_i = \frac{S_i}{L_i}$$

satisfying $Q_i L_i = S_i$. Thus a zero assembly residual is automatic after the target appears in the definition of Q; it does not identify a mechanism.

The repository correctly proves that the legacy cosine does not vanish on integer $d = 0, \ldots, 11$. The APD values are therefore finite there. Direct replay gives six positive and six negative multipliers, so Q_{APD} is not a positive diagonal measure, Markov channel or completely positive operation.

3.2 Global obstruction

For the legacy parameters,

$$\cos\left(\frac{\pi}{4}d + \frac{\pi}{6}\right) = 0 \quad \Longleftrightarrow \quad d = \frac{4}{3} + 4k.$$

At the first three nonnegative zeros the strict kernel equals approximately

$$0.3423917, quad 0.0189960, quad - 0.00563505,$$

and is nonzero. Therefore $K_{\mathrm{strict}}/K_{\mathrm{legacy}}$ has nonremovable poles.

Theorem 1 (APD global multiplicative no-go)**.** *No finite continuous multiplier on a real domain containing a legacy zero with nonzero strict target can satisfy* $K_{\mathrm{legacy}}Q = K_{\mathrm{strict}}$.

Around $d_{\mathrm{ref}} = 1$, the nearest real pole is at $4/3$, so the APD factor is a valid local analytic germ. Its moment equality is then a derivative of the identity used to define the ratio. This preserves P2363 exactly in its stated finite nonzero-domain/local-moment scope while forbidding promotion to a global physical completion operator.

An additive repair $K_{\mathrm{strict}} = K_{\mathrm{legacy}} + (K_{\mathrm{strict}} - K_{\mathrm{legacy}})$ is regular but imports the complete target difference. General nonlocal linear maps sending one finite vector to another are infinitely nonunique. Neither alternative supplies a strict source law.

4 Round III: ST1752–ST1766 — literal variation and gauge test

Suppressing only irrelevant tensor notation, the displayed terms contain

$$L = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4}\phi^4 - y\phi B + \frac{g}{2}A^2\phi^2 + \xi R\phi^2, \qquad B = \bar{\psi}\psi.$$

Using the equation sheet's convention $E_\phi = \partial(\partial L/\partial(\partial\phi)) - \partial L/\partial\phi$ gives

$$E_\phi = \Box\phi + m^2\phi + \lambda\phi^3 + yB - gA^2\phi - 2\xi R\phi.$$

The compact draft prints $-yB$. The relative Yukawa sign therefore disagrees with literal variation of its own displayed action.

For an Abelian gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu\alpha$, the infinitesimal variation of the seagull term is

$$\delta L = g\phi^2 A^\mu \partial_\mu \alpha = -g\alpha\, \partial_\mu(\phi^2 A^\mu) + \text{boundary},$$

which is not identically zero. The gauge EOM divergence inherits the same Ward defect.

A locally invariant scalar repair requires a complex charged field:

$$|D\Phi|^2 = |\partial\Phi|^2 + iqA_\mu(\Phi^*\partial^\mu\Phi - \Phi\partial^\mu\Phi^*) + q^2A^2|\Phi|^2.$$

The derivative current and seagull are mandatory and share the coefficient q. The current draft has the seagull with independent g_{eff} and no corresponding derivative current. Alternatively A may be read as a Proca field, but then local gauge/BRST language must be dropped or a Stueckelberg/ Higgs field added.

The quartic normalization also differs: $-\lambda\phi^4/4$ yields $\lambda\phi^3$, while the sector row $\lambda_4\phi^3/3!$ corresponds to a $\lambda_4\phi^4/4!$ convention. No map is exported, and the cubic term is absent from the displayed action.

Theorem 2 (single displayed action no-go)**.** *The current displayed* L_{total} *does not generate all current covariant EOM rows and is not a local gauge-invariant action in the declared field content.*

Reduced termwise models with zero recomposition residual remain internally valid as separate reduced actions. They do not prove the global claim.

5 Round IV: ST1767–ST1781 — moment-map falsification

The moment feature map is

$$(\omega, \phi, \beta, \eta; d_{\mathrm{ref}}) \longmapsto (c_0, c_1, c_2).$$

For the same strict kernel, moving d_{ref} from 0.5 to 1 to 2 changes the mass/ξ multiplier $1 + c_0$ from approximately 1.75170 to 1.46999 to 1.19204. The g multiplier $1 + c_2$ changes from 1.01823 to 1.23191 to 1.07847. No canonical reference length is exported.

Under $d' = ad$, the derivatives transform as

$$c_0'(1) = K(1/a), \quad c_1'(1) = a^{-1}K'(1/a), \quad c_2'(1) = \frac{1}{2}a^{-2}K''(1/a).$$

Thus the map is not coordinate-scale invariant.

At the nominal tuple, the 3×4 Jacobian has singular values

$$0.39633, quad 0.28037, quad 0.09566$$

and rank three. A local null direction exists. Fixing η to 1.79, 1.81 and 1.85, numerical root solves produce distinct (ω, ϕ, β) tuples with the same three moments to residuals between 10^{-15} and 10^{-12}, while their kernel values differ away from the reference point.

The five maps

$$m_{\text{eff}}^2 = m^2(1 + c_0), \quad \lambda_{\text{eff}} = \lambda(1 + c_1^2), \quad \ldots$$

retain five free base couplings. Wherever the factors are nonzero, any target effective coupling is obtained by dividing it by the factor. The map therefore does not predict absolute couplings. Squaring c_1 additionally deletes its sign.

A fourth normalized derivative raises the local parameter Jacobian to rank four with condition number about 10.8. This is a constructive mathematical repair of local parameter identifiability, not a physical source or scale.

6 Round V: ST1782–ST1796 — canonical action and propagator test

The strict finite generator defines the unavoidable Dirichlet action

$$S_A[f; J] = \frac{1}{2}f^T A f - J^T f.$$

Its variation is $Af = J$. Since $A\mathbf{1} = 0$, a solution requires $\mathbf{1}^T J = 0$ and is unique only modulo $f \mapsto f + c\mathbf{1}$. On the mean-zero space the Green operator is the Moore–Penrose inverse A^+. Adding $m^2 > 0$ gives the positive family

$$G_m = (A + m^2 I)^{-1},$$

but m is extra input.

In the frozen FIN matrix construction,

$$W = sI - A, qquad s = 1.660307278766099.$$

Suppose, even after normal-order subtraction, that

$$s - \lambda = \frac{a}{\lambda + m^2} - c$$

on the spectrum of A. Multiplication by $\lambda + m^2$ gives a nonconstant quadratic polynomial that cannot be constant at more than two distinct λ values. The strict A has seven distinct eigenvalues.

Theorem 3 (same-generator propagator obstruction)**.** *No constants a, m^2, c make the current W a normal-ordered resolvent of the same A on its full spectrum.*

Numerically, W has seven negative and five positive eigenvalues. It is invertible, but an action whose Euclidean covariance is W has Hessian W^{-1} and is unbounded below. Therefore W is currently an adjacency/ coupling matrix, not the positive Green covariance of the canonical A action. It could be a propagator of a different parent operator only if that operator and its stability structure are independently supplied.

The same A still generates heat, unitary and wave calculi. The Dirichlet action does not select which is physical. A noncommutative spectral action would additionally need an algebra, representation, Dirac grading/reality, cutoff function and dimensional cutoff.

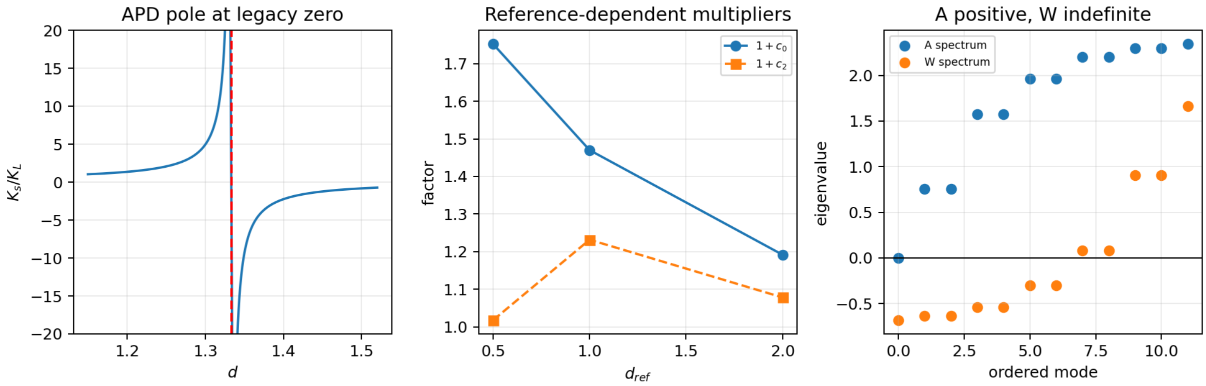

Figure 1: Left: the first nonremovable APD pole. Centre: reference dependence of selected coupling multipliers. Right: positive A spectrum versus indefinite W spectrum.

7 Round VI: ST1797–ST1811 — decisive verdict

7.1 What is proved, refuted and conditional

Object/claim	Status	Boundary
finite APD ratio on Z_{12}	[Proven]	automatic diagonal completion
global bounded/positive APD	[Refuted]	nonremovable poles/signs
P2363 local moment identity	[Proven local]	finite nonzero domain/germ
one displayed covariant action for all EOM	[Refuted]	term/sign/gauge defects
reduced/FRW normal forms	[Conditional]	named actions/backgrounds
three moments identify strict kernel	[Refuted]	rank-three four-parameter map
moment map predicts couplings	[Refuted]	free bases/reference choice
W is Green of same A	[Refuted]	spectral polynomial obstruction
finite Dirichlet form of A	[Proven]	dimensionless finite model
OA dual-dynamics protocols	[Conditional/certified]	selected state/channel/apparatus model
fundamental physical theory	[Not established]	decisive source-action rows absent

7.2 The decisive source–action gate

A future claim connecting FIN to unique physics must jointly supply:

G1. a globally regular, non-tautological strict source map;

G2. canonical refinement/continuum geometry and dimensional scale;

G3. one typed gauge/diffeomorphism-consistent action;

G4. literal variation regenerating every claimed EOM with residual tables;

G5. state, channel, clock, preparation, instrument and record;

G6. parameter values or sharp predictions not absorbable into free bases;

G7. a held-out physical forward model and falsifiable record.

Removing any row reproduces a known countermodel: APD quotient tautology, reference/unit torsors, gauge–Proca ambiguity, mixed-lineage EOM, A-only state nonuniqueness, free-coupling absorption or absence of physics evidence. The current strict repository has useful subcomponents, but zero complete rows passing all obligations of their class.

7.3 Route ranking

Route	Rigorous-success estimate	Physics decision power
publish finite Dirichlet/Markov/spectral FIN mathematics	0.90	0.25
derive refinement-compatible continuum Dirichlet form, scale and causal observable	0.35	0.80
derive gauge-covariant spectral/source action from independently sourced algebra	0.18	1.00
repair current APD–three-moment–L_{total} chain	0.03	0.70
conditional OA laboratory discrimination	0.45	0.55

These are strategic judgments, not probabilities derived from data.

8 Most important result and highest-value next direction

The most important result is the conjunction of independent obstructions:

$$\boxed{\text{current APD} \to \text{three moments} \to \text{free couplings} \to L_{\text{total}} \text{ is not a derivation of unique physics.}}$$

No single defect alone establishes this. The conclusion survives attempts to reinterpret APD locally, repair gauge invariance, move the reference point, add a fourth moment and reinterpret W as a propagator.

The deepest object surviving every falsification attempt is

$$\boxed{\text{a finite Dirichlet–Markov–spectral information geometry}}$$

together with exact contextual reductions and conditional operational channel models. It is mathematically coherent and potentially publishable, but it is not yet fundamental physics.

The single highest-value next theorem is:

> *derive or refute a globally regular, refinement-compatible, gauge-covariant source action directly from strict algebra/Dirichlet data, and regenerate every field equation by literal variation before fitting any Standard-Model or gravitational coefficient.*

The immediate falsification order is global regularity, Ward/gauge identity, Euler regeneration, refinement naturality, scale covariance and a nonabsorbable prediction. Failure at any row should stop the route. Repeating APD quotients, three-moment coupling maps or additional receiver-level SM/GR rows cannot resolve the established obstructions.

9 Final interpretation

> FIN is presently a rigorous finite spectral-information and Dirichlet/Markov framework with unusually strong internal no-go discipline and increasingly precise conditional operational tests. The current repository does not derive a globally regular bridge, a unique gauge-covariant action, physical units, state semantics or experimental law from the kernel. It therefore cannot presently be classified as a fundamental physical theory. Its most scientifically valuable next step is a single source–action theorem (or its no-go), not another phenomenological completion of the existing scaffold.

A Six-round ledger

Range	Target	Verdict
ST1722–ST1736	repository action/EOM map	mixed lineage and gauge defects
ST1737–ST1751	APD completion	finite/local valid; global physical no-go
ST1752–ST1766	literal variation	Yukawa sign, gauge and source mismatch
ST1767–ST1781	moment-to-coupling map	reference/rank/prediction no-go
ST1782–ST1796	canonical action/propagator	Dirichlet survives; W-Green refuted
ST1797–ST1811	synthesis	decisive seven-row source–action gate

B Reproducibility

The authoritative result sets run from `FIN_ST1722_ST1736_Results.json` through `FIN_ST1797_ST1811_Results.json`. Each contains fifteen programmes and packet SHA-256 references. The combined test `test_fin_st1722_st1811.py` verifies all ninety packets, APD poles/signs, literal variation, moment rank, propagator spectrum, decisive gate, figure and nonpromotion boundaries.

C Nonclaims

This report does not prove that no future FIN extension can become physical. It does not invalidate the finite strict kernel, Dirichlet form, spectral theorems, reduced actions or conditional OA tests in their stated scopes. It does not complete legacy role transfer, QW-2191, units, Standard Model, gravity, L_{total} or Theory of Everything. All future FIN PDFs remain English.

References

[1] M. Reed and B. Simon, *Methods of Modern Mathematical Physics I*, Academic Press.

[2] M. Peskin and D. Schroeder, *An Introduction to Quantum Field Theory*, Addison–Wesley.

[3] A. Connes and M. Marcolli, *Noncommutative Geometry, Quantum Fields and Motives*, AMS.